

An Introduction to Quantum Computing

Quantum computing is a model of computation that exploits quantum-mechanical phenomena — superposition, entanglement, and interference — to process information in ways a classical computer cannot efficiently replicate. Where a classical bit is strictly 0 or 1, a quantum bit can occupy a combination of both states at once, enabling certain algorithms to explore an exponentially large space in parallel.

Qubits and Superposition

A qubit is the fundamental unit of quantum information. Unlike a classical bit, a qubit's state is a vector in a two-dimensional complex space, written as a combination of the basis states $|0\rangle$ and $|1\rangle$. This property, called superposition, means a register of n qubits can represent 2^n amplitudes simultaneously. Measurement collapses the superposition to a single classical outcome with a probability set by the amplitudes.

Entanglement

Entanglement is a uniquely quantum correlation between qubits such that the state of the whole system cannot be described as the product of individual qubit states. Measuring one entangled qubit instantly constrains the outcome of its partner, regardless of distance. Entanglement is the key resource behind quantum teleportation, superdense coding, and the speedups of many quantum algorithms.

Quantum Algorithms

Quantum algorithms are procedures that orchestrate superposition and interference to make correct answers more probable upon measurement. The two best-known examples are Shor's algorithm and Grover's algorithm, which target very different problems.

Shor's Algorithm

Shor's algorithm factors large integers in polynomial time by reducing factoring to period-finding, solved efficiently with the quantum Fourier transform. Because the security of RSA encryption rests on the hardness of factoring, a large fault-tolerant quantum computer running Shor's algorithm would break widely deployed public-key cryptography, motivating the field of post-quantum cryptography.

Grover's Algorithm

Grover's algorithm searches an unstructured database of N items in roughly the square root of N steps, a quadratic speedup over classical brute force. It works by amplitude amplification: repeatedly inverting the amplitude of the target state about the average, nudging probability toward the correct answer before measurement.

Quantum Error Correction

Physical qubits are fragile and decohere quickly, so practical machines need quantum error correction. Codes such as the surface code spread one logical qubit across many physical qubits, detecting and correcting errors without directly measuring (and thus destroying) the encoded information. Fault tolerance requires error rates below a threshold and a large overhead of physical qubits per logical qubit.

Hardware Approaches

Several physical platforms compete to build qubits: superconducting circuits (used by Google and IBM), trapped ions (IonQ, Quantinuum), photonic systems, and neutral atoms. Each trades off gate speed, connectivity, coherence time, and manufacturability. No single approach has yet achieved large-scale fault tolerance, and the engineering race remains wide open.